Grade: Check

Nick Ryan | Statistics and Data Analysis | Due 2/24/2026

Assignment #2

1. Compute the mean and SD for the stock price

Price (x)	Probability f(x)	x*f(x)	x^2*f(x)
20	0.05	1	20
21	0.06	1.26	26.46
22	0.07	1.54	33.88
23	0.08	1.84	42.32
24	0.09	2.16	51.84
25	0.12	3	75
26	0.15	3.9	101.4
27	0.13	3.51	94.77
28	0.10	2.8	78.4
29	0.10	2.9	84.1
30	0.05	1.5	45
	Mean	25.41	653.17
	Variance		7.50
	Standard Deviation		2.74

✓ Q1

2. What is the probability that the stock price will fall within 1 SD of the mean in one year

✓ Q2a

   a. Mean Price - $25.41. Standard Deviation = 2.74
      $25.41±2.74 = $22.67 and $28.15
      Possible prices in this range = $23, $24, $25, $26, $27, and $28
      Probability of these = .08 + .09 + .12 + .15 + .13 + .10 = .67 or 67%
   b. What is the probability that the stock price will fall within 2 SD of the mean in one year
      Mean Price - $25.41. Standard Deviation = 2.74
      $25.41±2(2.74) = $19.93 and $30.89
      Possible prices in this range = $20, $21, $22, $23, $24, $25, $26, $27, $28, $29, and $30

✓ Q2b

Probability of these = .05 + .06 + .07 + .08 + .09 + .12 + .15 + .13 + .10 + .10 + .05= 1.0 or 100%

3. Derive the distribution of the option value, ignoring interest and commissions

x Q3: Needs revision.

The option value ranges from -5 (at a $20 stock price) to +5 (at a $30 stock price)

4. What is the probability that the option will have a positive value

✓ Q4

The option will only have a positive value for stock prices greater than $25. The probability of a stock price greater than $25 is .15 + .13 + .1 + .1 + .05 = .53 or 53%

5. Compute the mean and SD of the value of the option

Mean (ignoring all non-positive values as the option is worthless) = 1.36

Standard Deviation = 1.60

Option Value	Value * Probability	Value^2 * probability
1	0.15	0.15
2	✓ Q5 0.26	0.52
3	0.3	0.9
4	0.4	1.6
5	0.25	1.25
Mean	1.36	4.42
	Variance	2.57
	Standard Deviation	1.60